

Image Processing

Introduction:

motivation, overview and logistics

Francesco Solera

October 1st, 2026

My background

Francesco Solera

Academic

- Master in Computer Engineering @ UniMoRe
- Phd in Computer Vision in 2016 @ UniMoRe
- Brief period as Visiting Researcher at Duke University

Work experience:

- Since 2017 in Deep Vision Consulting, now a partner.
- My area of expertise is Deep Learning applied to Computer Vision.

My company background



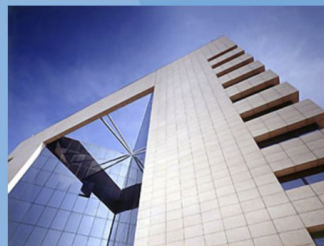
Deep Vision Consulting

<https://deepvisionconsulting.com/>

Specializes in the development of advanced computer vision and artificial intelligence software, algorithms, and intellectual properties.

Topics:

AI, machine learning and deep learning, at-the-edge computing, embedded systems, data simulations and robotics.



Computer Vision is wide



IMAGE ACQUISITION

sensor
lens
illumination
filters

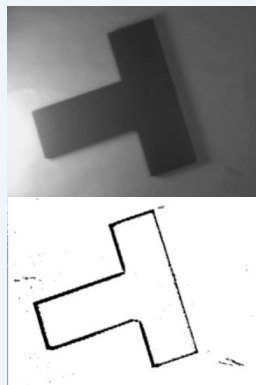
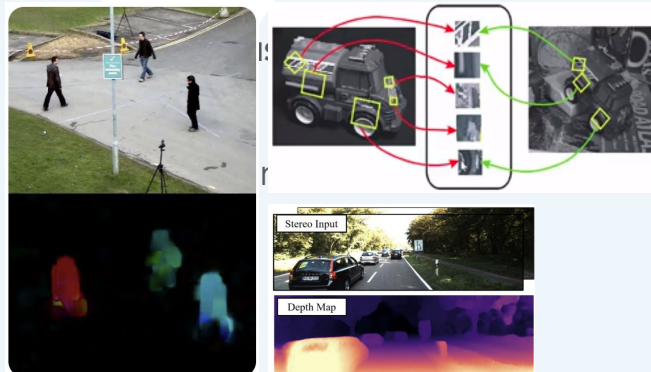


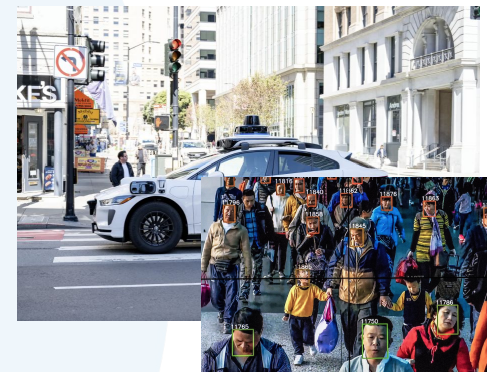
IMAGE PROCESSING

pointwise ops
filters
edges & regions
labeling
colors
calibration
measurement



MID COMPUTER VISION

detection
tracking
segmentation
motion (and video analysis)
3D
local feature descriptors
machine learning



ADVANCED COMPUTER VISION

all previous boosted:

- more data
- larger models
- more compute

face recognition
self driving cars

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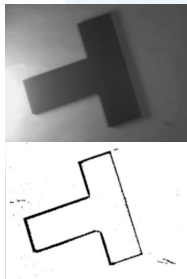
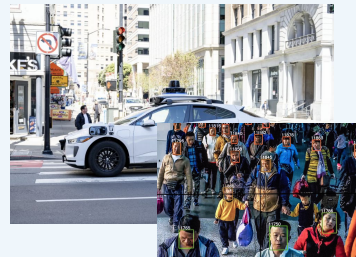
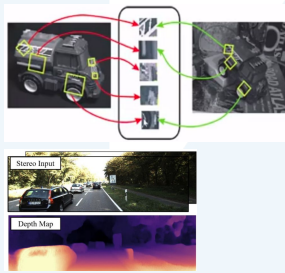


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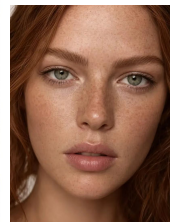


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FRONTIER COMPUTER VISION
- image / video / 3D generation
- zero shot promptable models
- physical AI
- world models to train robots in simulation



... research in computer vision is at its peak!

Computer Vision is wide



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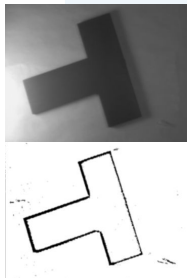
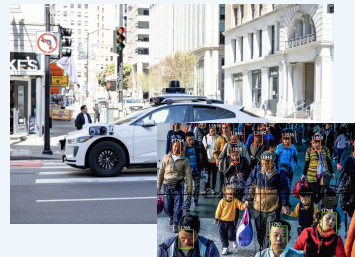
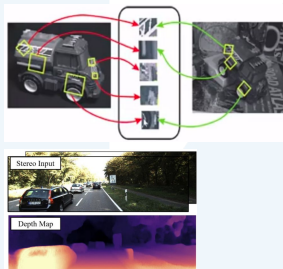


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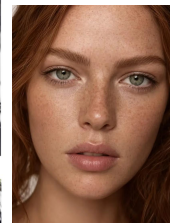


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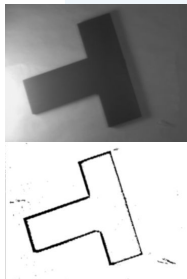
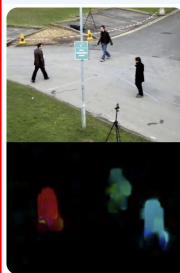
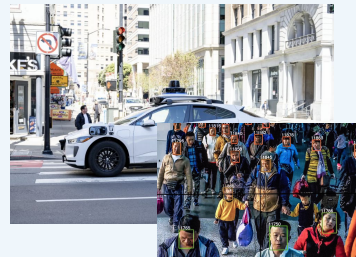
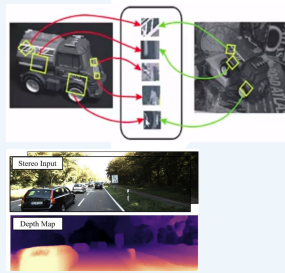


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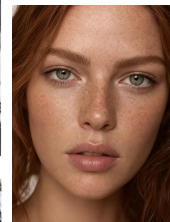


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YES! And people who only know (a bit) of the most modern CV are missing very useful skills! Knowing IA and IP will make a real difference if you want to apply CV (also more advanced one) in industry!

Computer Vision is wide



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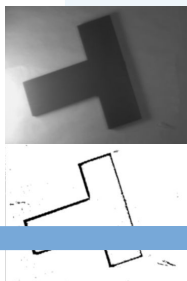
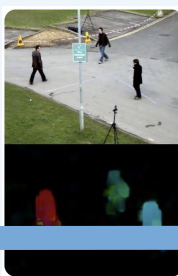
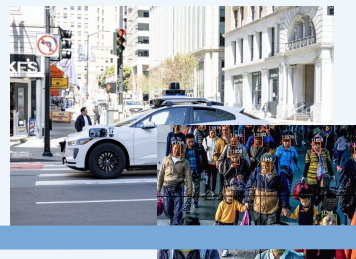
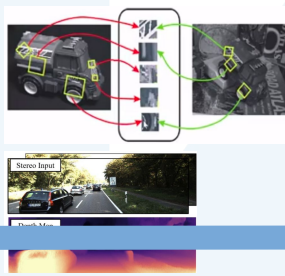


IMAGE PROCESSING



MID COMPUTER VISION



ADVANCED COMPUTER VISION



FRONTIER COMPUTER VISION



You should try hard to solve a problem from left to right, because as you move on the right:

- you need more data or be prepared to lose control
- you lose interpretability
- you need more computation
 - larger hardware (maybe through cloud access)
 - more processing time
 - more power

Only move to the right if you see a clear better complexity tradeoff in doing so. Most of the time you can also split your problem into smaller ones and solve different parts at different semantic levels.

Syllabus of my lectures

IUX

01 / 10 / 2026: image formation I: thin lenses and optics tradeoffs

IUX

13 / 10 / 2026: image formation II: from continuous light to a digital image

How IUX pages look like




IMAGE PROCESSING
INGEGNERIA INFORMATICA (MN) - 2026/2027

Show topics ▾

IMAGE FORMATION

Thin lenses

STEP-BY-STEP TUTORIAL
Balance the equation
 Can you predict the third term? Keep f fixed and bring the object closer; watch d_i grow and the image enlarge. Then change what is fixed to test how any two terms determine the third.

< Previous
 Step 3 of 8
 Next >

Parameters

Focal length f
 distance between focal point F and optical center of the lens
☒ Fix
 40 mm

Object distance d_o
 distance between the object and the optical center of the lens
☐ Fix
 120 mm

Image distance d_i
 distance between the image plane and the optical center of the lens
☐ Fix
 60 mm

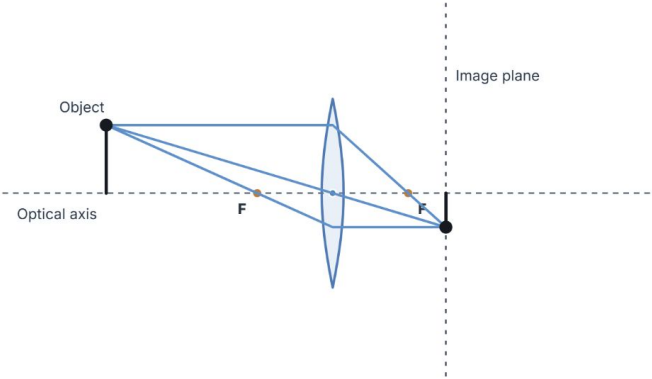
Object height h_o
 perpendicular distance of the object from the

THIN-LENS EQUATION

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

IMAGE PLANE FRONT VIEW


Trace light rays
 Parallel
 Central
 Focal



Syllabus of my lectures

- IUX 01 / 10 / 2026: image formation I: thin lenses and optics tradeoffs
- IUX 13 / 10 / 2026: image formation II: from continuous light to a digital image
- LAB 15 / 10 / 2026: intro to opencv: basic camera, image and video controls
- LAB 20 / 10 / 2026: from edges to contours: intro to thresholding & morphology
- LAB 22 / 10 / 2026: looking for objects I: labeling, shapes and template matching
- LAB 29 / 10 / 2026: looking for objects II: dealing with noisy data
- LAB 12 / 11 / 2026: camera calibration, undistortion and planar homographies

How LAB lectures are structured

PART I - approx. 120 mins
theory



PART II - approx. 60 mins
lab intro + small practice



PART III
additional practice points

Part I and II will fall into the 3 hours weekly lectures. AI is not allowed during the lectures.

Part III is totally up to you. Consider it a chance to learn more, and more deeply.

Solutions will always be provided for all exercises. It's up to you not to look at them too early.

For any doubt about both theory or lab topics and solutions, drop me an email. It may be helpful to others too.

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- LAB 20 / 10 / 2026: from edges to contours: intro to thresholding & morphology
- LAB 22 / 10 / 2026: looking for objects I: labeling, shapes and template matching
- EXT 27 / 10 / 2026: [MODENA] visit to Deep Vision Consulting & AImageLab
- LAB 29 / 10 / 2026: looking for objects II: dealing with noisy data
- LAB 12 / 11 / 2026: camera calibration, undistortion and planar homographies
- EXT 03 / 12 / 2026: [MANTOVA] external visit at Opto Engineering

Other lectures and topics will be covered by prof. Sangineto.

External visits



1) Deep Vision Consulting

Based in Modena, specialized in advanced computer vision algorithms to solve real world problems.

2) UniMoRe Computer Vision Lab

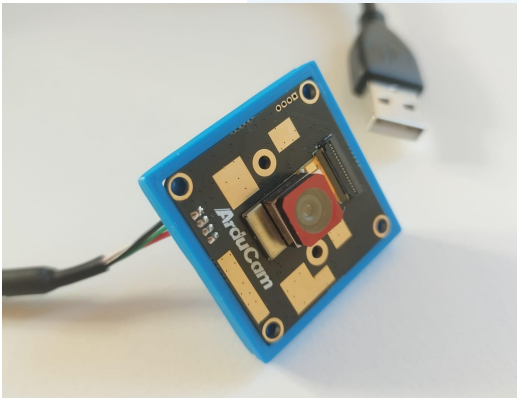
Research lab tackling frontier research questions and teaching advanced computer vision master courses at UniMoRe.

3) Opto Engineering

Excellent company based in Mantova, focused on high quality lenses and cameras.

I will follow up by email with more details and to collect participation that is of course optional in all 3 cases.

Lab Kit sponsored by EBV Elektronik & Arducam



EBV Elektronik, an Avnet company, is a leading European specialist in semiconductor distribution, providing:

- electronic components,
- technical expertise, and
- design support for innovative embedded and imaging applications.

Arducam develops and manufactures custom camera modules and imaging solutions for computer and embedded vision applications.

We gratefully acknowledge EBV's and Arducam's support of this course through the donation of a webcam to every student for the laboratory sessions.

Questions?

mail me @

francesco.solera@deepvisionconsulting.com

francesco.solera@unimore.it